# AXEL: ZERO — Master Build Prompt (Validation Prototype)

**For:** Fresh Claude Code session, web/Next.js project. **Goal:** Ship a playable cascading-cluster slot roguelite prototype in 1–2 weeks to validate K-factor before committing to the full AXEL build. **Philosophy:** Speed > polish. The cascade clip moment + daily seed leaderboard are the validation. Ship ugly, ship fast.

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## 1. Project Overview

You are building **AXEL: ZERO** — a web-playable, emoji-symbol, cascading-cluster slot roguelite.

**Why this will work (wave-riding context):**

- CloverPit shipped 100k copies in 1 day (slot roguelite zeitgeist peaking).
- Balatro proved the "gambling-mechanic-as-deep-strategy" pattern.
- Moon Princess 100's cascade-cluster mechanic has NOT yet been brought to the roguelite frame.
- Sub-genre window: 6–12 weeks before saturation.

**What we're copying (legally safe):** cluster pays (4+ touching symbols), cascading avalanche on win, multiplier wild that ramps within a spin, grid-clear → free spins. **What we're NOT touching:** Moon Princess naming/visuals, "Trinity" naming, princess/Sailor-Moon aesthetic.

## 2. Context Window Management

Work in segmented sessions. After each phase, run `/clear` and start fresh from the next phase's checklist.

- **Session 1:** Core cascade engine (Phase 1)
- **Session 2:** Run loop + items (Phase 2)
- **Session 3:** Sharing + leaderboard (Phase 3)
- **Session 4:** Polish + deploy (Phase 4)

## 3. Tech Stack

- **Frontend:** Next.js 14 (app router) + TypeScript + Tailwind
- **Styling:** CSS grid for the 5×5 slot board, CSS animations for cascade

- **Backend:** Supabase (shared instance `sxoztxicojvfwquhtmpk`) — table for daily seeds + leaderboard
- **Share cards:** Vercel OG image generation (`@vercel/og`)
- **Deploy:** Vercel
- **No paid art:** Emoji symbols only

## 4. Reusable Code Reference

- `forge/blox-fruit-finder/` — pattern for emoji-quiz Next.js + Supabase + share-card
- `forge/viral/templates/` — existing scaffolds
- Shared Supabase patterns: see `memory/project_shared_supabase_instance.md`

## 5. Segmented TODO List

### PHASE 1 — CORE CASCADE ENGINE (Session 1, ~3 days)

- [ ] Next.js + TS + Tailwind scaffold
- [ ] 5×5 board state (2D array of `Symbol`)
- [ ] Symbol pool: 6–8 emoji symbols with weights
- [ ] `spin()` — fills board with weighted random symbols
- [ ] `findClusters(board)` — flood-fill, returns all clusters of 4+ touching same-symbol
- [ ] `resolveCascade(board)` — clears winning clusters, cascades remaining symbols down, fills empty cells, repeats until no wins
- [ ] Win calc: per-symbol payout × cluster size × current multiplier
- [ ] **Multiplier meter ("Chain Charge"):** +1 per cascade step in a spin, resets between spins, max 20×
- [ ] **Grid-clear → 3 free spins** with multiplier carryover
- [ ] CSS animations for clear (fade), cascade (drop), and multiplier-tick

### PHASE 2 — RUN LOOP + ITEMS (Session 2, ~3 days)

- [ ] Ante system: balance target rises each ante, lose if miss
- [ ] Shop between antes: pick 1 of 3 items
- [ ] ~12 items, each modifying symbol weights, cascade behavior, or multiplier ramp. Examples:
  - "Lucky Cog" — +20% weight on lowest-value symbol
  - "Resonator" — multiplier starts at 2× this spin
  - "Greaser" — cascades that clear ≥ 3 clusters double next spin's payout
- [ ] Run state: ante, coins, items, current spin

- [ ] Game over screen with seed, items collected, max-multiplier hit
- [ ] **CRITICAL:** Tune RTP per ante to be **sub-casino** (~85–92% expected, with high variance) — players FEEL like they're beating it, math is honest

### PHASE 3 — SHARING + LEADERBOARD (Session 3, ~3 days)

- [ ] Daily seed: deterministic PRNG keyed on `YYYY-MM-DD`
- [ ] Supabase table: `axel_zero_runs(id, date, score, max_multiplier, items, seed, created_at)`
- [ ] Submit run on game-over
- [ ] Leaderboard page: today's top 100
- [ ] **Auto-share card via @vercel/og:** PNG with score, max multiplier, items, seed. Single "Share" button copies URL.
- [ ] Twitter/X meta tags so URL unfurls beautifully

### PHASE 4 — POLISH + DEPLOY (Session 4, ~2 days)

- [ ] Sound: free pack from freesound — cascade clink, multiplier ramp, grid-clear fanfare
- [ ] Mobile responsive (touch-friendly)
- [ ] Onboarding: 30s tutorial overlay first play
- [ ] Deploy to Vercel, custom domain (e.g. axel-zero.vercel.app or axel.game)
- [ ] Plausible/Umami for K-factor tracking: shares, returns, daily uniques

## 6. Game-Specific Details

**Symbol pool (emoji-only, easy to swap later):**

- 🔧 (common, 1×)
- 🪙 (common, 2×)
- ⚙️ (uncommon, 3×)
- 💎 (rare, 8×)
- 🔥 (wild — multiplier)
- 🎰 (scatter — grid-clear bonus)

**Cluster pays table:**

- 4 touching = 1× base
- 5–7 = 2×
- 8–11 = 5×
- 12+ = 20×

**Viral hook:** Big cascade clears → multiplier hits 10×+ → grid clears → free spins burst. THIS is the moment the share card screenshot captures.

## 7. Critical Reminders

- ❌ **Do not** call this game "Moon Princess Cluster" or anything princess/anime-themed
- ❌ **Do not** name the multiplier meter "Trinity"
- ❌ **Do not** add real-money or premium-IAP framing in this prototype
- ✅ Lean hard on the **cascade visual moment** — this is your TikTok-clip generator
- ✅ Auto-share card MUST be one-tap and visually striking
- ✅ Ship in 2 weeks max. If polish creeps, cut content, not the loop.

## 8. Launch Checklist

- ☐ Deploy to Vercel
- ☐ Buy domain (axel.game or axel-zero.vercel.app fine for v1)
- ☐ Post on r/incremental_games, r/roguelikes, r/WebGames
- ☐ Tweet with a 6-second cascade clip — screen-record one big cascade
- ☐ Hand-share to 5 streamers who play Balatro/CloverPit
- ☐ **Kill criteria:** if 14-day K-factor < 0.3, kill and pivot to AXEL: Resonance directly
- ☐ **Continue criteria:** K > 0.5 → start AXEL: Resonance master prompt